Week 9: The Black–Scholes Model from First Principles

Connecting Global Strategy with Financial Decisions

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Learning Objectives

- Explain why continuous compounding produces the number e.
- Distinguish a normal distribution from a lognormal distribution.
- Interpret Brownian motion, dt, dW, and a stochastic differential equation.
- Derive Itô's lemma from a second-order Taylor expansion.
- Construct the Black–Scholes hedge and derive the pricing PDE.
- Derive the European call formula and interpret d_1 and d_2.

Lecture Roadmap

9.1 Exponential Growth and the Number e

9.2 Probability Foundations

9.3 Normal and Lognormal Distributions

9.4 Random Walks and Brownian Motion

9.5 Stochastic Differential Notation

9.6 Geometric Brownian Motion

Lecture Roadmap (continued)

9.7 Itô's Lemma

9.8 No-Arbitrage and Replication

9.9 The Black–Scholes PDE

9.10 Solving the PDE

9.11 The Black–Scholes Formula

9.12 Interpretation, Extensions, and Limits

Teaching Plan

- First part: mathematical foundations with numerical examples.
- Second part: complete derivation of the option-pricing model.
- Final 30 minutes: guided exercises followed by solutions.

Common mistake

Students should explain each symbol before substituting numbers.

9.1 Exponential Growth and the Number e

Why continuous time naturally creates an exponential function

Simple Interest

- Simple interest applies the rate only to the original principal.
- For principal P, annual rate r, and time t, the future value is
- $$FV = P(1 + rt).$$
- The interest earned during one period does not itself earn interest.

Compound Interest

- Compounding reinvests interest after every compounding date.
- If the year contains m compounding periods,
- $$FV_m = P\left(1 + \frac{r}{m}\right)^{mt}.$$
- Increasing m makes compounding more frequent.

The Limit that Defines e

Set $r = 1$, $t = 1$, and let the number of compounding periods increase:

$$\left(1 + \frac{1}{m}\right)^m.$$

m	1	2	4	12	365
Value	2.0000	2.2500	2.4414	2.6130	2.7146

The sequence approaches

$$e = \lim_{m \to \infty} \left(1 + \frac{1}{m}\right)^m \approx 2.71828.$$

Why e Appears in Finance

- Continuous compounding takes the limit as the compounding interval approaches zero.
- $$\lim_{m\to\infty} P\left(1+\frac{r}{m}\right)^{mt} = Pe^{rt}.$$
- The exponential function is the unique growth function whose instantaneous growth rate is proportional to its current level.
- $$\frac{d}{dt}e^{rt} = re^{rt}.$$

Exponential Growth as a Differential Equation

Suppose value grows at instantaneous rate r:

$$\frac{dV_t}{dt} = rV_t.$$

Separate variables:

$$\frac{dV_t}{V_t} = r\,dt.$$

Integrate from 0 to t:

$$\ln V_t - \ln V_0 = rt.$$

Exponentiating gives

$$V_t = V_0 e^{rt}.$$

Discounting with Continuous Compounding

If one dollar grows to e^{rT} dollars at time T, then one dollar received at time T has present value

$$PV = e^{-rT}.$$

For a certain payment K at time T,

$$PV(K) = Ke^{-rT}.$$

Finance interpretation

The term e^{-rT} appears in Black–Scholes because the strike payment occurs at expiration and must be discounted continuously.

Logarithms Undo Exponentials

- If $y = e^x$, then $x = \ln y$.
- $\ln(ab) = \ln a + \ln b$ converts multiplication into addition.
- $\ln(a^q) = q \ln a$ brings powers down as coefficients.
- These properties make continuously compounded returns additive through time.

Example

If $S_T = S_0 e^R$, then $R = \ln(S_T/S_0)$.

9.2 Probability Foundations

The minimum probability language needed for stochastic calculus

Random Variables

- A random variable assigns a number to each possible outcome.
- A discrete random variable takes countable values.
- A continuous random variable can take any value in an interval.
- A stock return is uncertain today and becomes an observed number later.

Expected Value

For discrete outcomes,

$$\mathbb{E}[X] = \sum_i p_i x_i.$$

For a continuous variable with density $f(x)$,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\, dx.$$

Expected value is a probability-weighted average, not a guaranteed outcome.

Variance and Standard Deviation

Variance measures average squared deviation from the mean:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Standard deviation uses the original unit:

$$\sigma_X = \sqrt{\mathrm{Var}(X)}.$$

In an option model, volatility is the annualized standard deviation of returns.

Conditional Expectation

- $\mathbb{E}[X \mid \mathcal{F}_t]$ means the expected value of X using information available at time t.
- The information set grows as prices and news arrive.
- Option value at time t depends on the distribution of future payoffs conditional on current information.

Notation

$\mathcal{F}_t$ represents everything known by time t.

Independence

- Two increments are independent when observing one provides no information about the other.
- For independent X and Y, $\mathrm{Cov}(X, Y) = 0$.
- Brownian motion assumes independent increments over nonoverlapping time intervals.

Common mistake

Zero covariance does not always imply independence. Joint normality makes the implication valid.

The Central Limit Idea

- A long-period return combines many small shocks.
- Under suitable conditions, a standardized sum of independent small shocks approaches a normal distribution.
- This limiting idea motivates normal increments in Brownian motion.
- Market returns are not exactly normal, especially in crises.

9.3 Normal and Lognormal Distributions

Why the model applies normality to log returns rather than prices

The Standard Normal Distribution

A standard normal variable Z has density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

Its cumulative distribution function is

$$N(z) = \Pr(Z \leq z) = \int_{-\infty}^{z} \phi(u)\, du.$$

The function $N(\cdot)$ appears in the Black–Scholes formula.

Why the Normal Density Contains e

A bell-shaped density with derivative proportional to $-z$ times itself satisfies

$$\phi'(z) = -z\phi(z).$$

Solving this differential equation gives

$$\phi(z) = Ce^{-z^2/2}.$$

The constant $C = 1/\sqrt{2\pi}$ makes the total area equal to one.

Normal Variables Can Be Negative

- A normal variable ranges from $-\infty$ to $+\infty$.
- Modeling the stock price itself as normal assigns positive probability to a negative price.
- Limited liability prevents an ordinary share price from falling below zero.
- The model therefore places a normal distribution on the logarithm of price.

Definition of a Lognormal Variable

If

$$Y = \ln X \sim N(m, v),$$

then

$$X = e^Y$$

has a lognormal distribution. Because the exponential function is positive,

$$X > 0.$$

The distribution is right-skewed and allows large upside outcomes.

Mean of a Lognormal Variable

If $Y \sim N(m, v)$ and $X = e^Y$, then

$$\mathbb{E}[X] = e^{m+v/2}.$$

The term $v/2$ matters. In the stock model, this term creates the $-\sigma^2/2$ adjustment in expected log return.

Price Ratios and Log Returns

The continuously compounded return is

$$R_{t,T} = \ln\left(\frac{S_T}{S_t}\right).$$

If log return is normal, then

$$\frac{S_T}{S_t} = e^{R_{t,T}}$$

is lognormal and remains positive.

Why Log Returns Are Convenient

- Additivity: $\ln(S_2/S_0) = \ln(S_1/S_0) + \ln(S_2/S_1)$.
- Positivity: exponentiating a real number gives a positive price.
- Scale: percentage changes matter more than equal dollar changes across different price levels.
- Aggregation: sums of many small log-return shocks support a normal approximation.

Modeling Choice, Not a Natural Law

- Actual returns show jumps, volatility clustering, skewness, and heavy tails.
- The lognormal assumption creates a tractable benchmark.
- Practitioners often use implied volatility, stochastic volatility, or jump models when the benchmark is inadequate.

Common mistake

The model does not claim that every observed return is normally distributed.

9.4 Random Walks and Brownian Motion

From many small shocks to a continuous random path

A Discrete Random Walk

Let each one-period shock be $\varepsilon_i \in \{-1, +1\}$ with equal probability. Define

$$X_n = \sum_{i=1}^{n} \varepsilon_i.$$

Then

$$\mathbb{E}[X_n] = 0, \qquad \mathrm{Var}(X_n) = n.$$

The uncertainty grows with the number of steps.

Scaling a Random Walk

Divide a fixed horizon T into n intervals of length

$$\Delta t = T/n.$$

Scale each shock by $\sqrt{\Delta t}$:

$$\Delta W_i = \sqrt{\Delta t}\,\varepsilon_i.$$

Then

$$\text{Var}\left(\sum_{i=1}^{n} \Delta W_i\right) = n\Delta t = T.$$

Why the Shock Scales with $\sqrt{dt}$

If each shock scaled with dt, total variance would be

$$n(dt)^2 = T\,dt \to 0.$$

All randomness would disappear. If each shock had fixed variance, total variance would explode. Scaling by $\sqrt{dt}$ gives

$$n(\sqrt{dt})^2 = n\,dt = T,$$

a finite nonzero variance.

Brownian Motion

- $W_0 = 0$.
- Increments over nonoverlapping intervals are independent.
- $W_{t+h} - W_t \sim N(0, h)$.
- Sample paths are continuous with probability one.
- The path is continuous but almost surely nowhere differentiable.

Stationary Increments

For any t and $h > 0$,

$$W_{t+h} - W_t \sim N(0, h).$$

Only the interval length h matters. The calendar starting point t does not change the increment distribution.

$$\mathbb{E}[W_{t+h} - W_t] = 0, \qquad \mathrm{Var}(W_{t+h} - W_t) = h.$$

Why Brownian Motion Appears in Finance

- Price changes reflect many small pieces of information.
- The aggregate of many small shocks suggests a continuous Gaussian limit.
- Independent increments create a mathematically manageable benchmark.
- The model provides continuous trading and closed-form valuation.

Common mistake

This explanation motivates the assumption. It does not prove that real prices must follow Brownian motion.

Brownian Paths Are Rough

- Over a short interval, the typical increment has size $\sqrt{dt}$.
- The ratio dW/dt has typical size $1/\sqrt{dt}$ and becomes unstable as $dt \to 0$.
- Ordinary derivatives do not exist along a Brownian path.
- Stochastic calculus defines integrals directly rather than differentiating the path.

Quadratic Variation

For a partition $0 = t_0 < \cdots < t_n = T$,

$$\sum_{i=1}^{n} (W_{t_i} - W_{t_{i-1}})^2 \to T.$$

This nonzero limit is Brownian quadratic variation. In differential shorthand,

$$(dW)^2 = dt.$$

This identity creates the second-derivative term in Itô's lemma.

9.5 Stochastic Differential Notation

What dt, dW, and dS actually mean

Meaning of dt

- dt denotes a very small time interval in differential notation.
- In a rigorous construction, equations describe limits of sums as the largest interval approaches zero.
- A deterministic change proportional to dt accumulates linearly with time.

Meaning of dW

Over a short interval dt,

$$dW \sim N(0, dt).$$

Equivalently,

$$dW = \sqrt{dt}\, Z, \qquad Z \sim N(0, 1).$$

Therefore dW has order $\sqrt{dt}$, which is larger than dt over a very short interval.

The Itô Multiplication Rules

The orders of small quantities imply

$$(dt)^2 = 0, \qquad dt\, dW = 0, \qquad (dW)^2 = dt.$$

The first two terms vanish relative to dt. The last term remains because Brownian motion has nonzero quadratic variation.

Meaning of a Stochastic Differential Equation

The notation

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t$$

is shorthand for the integral equation

$$X_t = X_0 + \int_0^t a(X_s, s)ds + \int_0^t b(X_s, s)dW_s.$$

The first integral is ordinary. The second is an Itô stochastic integral.

Drift and Diffusion

- $a(X_t, t)dt$ is the conditional expected change over a short interval.
- $b(X_t, t)dW_t$ is the unpredictable shock.
- The drift has order dt. The diffusion has order $\sqrt{dt}$.
- At very high frequency, random variation dominates the local path.

Euler Discretization

For a small finite step Δt,

$$X_{t+\Delta t} \approx X_t + a(X_t, t)\Delta t + b(X_t, t)\sqrt{\Delta t}\, Z.$$

This approximation connects stochastic notation to simulation and spreadsheets.

9.6 Geometric Brownian Motion

The stock-price process used in Black–Scholes

The Stock-Price SDE

Black–Scholes assumes

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

Dividing by S_t gives

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t.$$

μ is the instantaneous expected return and σ is volatility.

Why the Model Is Multiplicative

- The dollar volatility σS_t rises with the price level.
- The model describes proportional rather than fixed dollar changes.
- The same percentage shock has a larger dollar effect on a more valuable share.
- The resulting price remains positive.

A Tempting but Incorrect Log Step

One might incorrectly write

$$d \ln S_t = \frac{dS_t}{S_t}.$$

This ordinary-calculus rule misses Brownian quadratic variation. The correct expression contains an additional term:

$$d \ln S_t = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dW_t.$$

Solution of Geometric Brownian Motion

Integrate the correct log equation from 0 to T:

$$\ln \frac{S_T}{S_0} = \left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma W_T.$$

Exponentiating,

$$S_T = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma W_T\right].$$

Distribution of the Future Price

Because $W_T \sim N(0, T)$,

$$\ln S_T \sim N\left(\ln S_0 + \left(\mu - \frac{1}{2}\sigma^2\right) T,\ \sigma^2 T\right).$$

Therefore S_T is lognormal. Its expectation is

$$\mathbb{E}[S_T] = S_0 e^{\mu T}.$$

9.7 Itô's Lemma

The stochastic version of the chain rule

Ordinary Taylor Expansion

For a smooth function $f(x,t)$,

$$df \approx f_t dt + f_x dx + \frac{1}{2} f_{xx} (dx)^2 + f_{xt} dx dt + \frac{1}{2} f_{tt} (dt)^2 + \cdots$$

In ordinary calculus, $(dx)^2$ usually vanishes faster than dt. Brownian motion changes this ordering.

Insert a Stochastic Change

Let

$$dX = a\,dt + b\,dW.$$

Then

$$(dX)^2 = a^2(dt)^2 + 2ab\,dt\,dW + b^2(dW)^2.$$

Using the Itô rules,

$$(dX)^2 = b^2 dt.$$

Terms That Survive

In the Taylor expansion:

$$(dt)^2 \text{ has order } dt^2 \to 0,$$

$$dt\, dW \text{ has order } dt^{3/2} \to 0,$$

$$(dW)^2 \text{ has order } dt \text{ and remains.}$$

Therefore the second derivative with respect to x cannot be discarded.

Itô's Lemma in One Dimension

If

$$dX = a(X,t)dt + b(X,t)dW$$

and $f = f(X,t)$, then

$$df = \left(f_t + af_x + \frac{1}{2}b^2 f_{xx}\right)dt + bf_x dW.$$

Key difference from ordinary calculus

The correction term is $\frac{1}{2}b^2 f_{xx}dt$.

Why the Factor 1/2 Appears

- The coefficient comes directly from the second-order Taylor term.
- Taylor expansion contains $\frac{1}{2} f_{xx} (dX)^2$.
- Brownian quadratic variation converts $(dX)^2$ into $b^2 dt$.
- The surviving term becomes $\frac{1}{2} b^2 f_{xx} dt$.

Example: $f(X) = X^2$

Here $f_x = 2X$ and $f_{xx} = 2$. If $dX = a\,dt + b\,dW$, then

$$d(X^2) = 2X\,dX + b^2 dt.$$

Substituting dX,

$$d(X^2) = (2aX + b^2)dt + 2bX\,dW.$$

Ordinary calculus would miss the $b^2 dt$ term.

Example: $f(X) = \ln X$

For $f(X) = \ln X$,

$$f_x = \frac{1}{X}, \qquad f_{xx} = -\frac{1}{X^2}.$$

With $dS = \mu S dt + \sigma S dW$,

$$d \ln S = \frac{1}{S} dS - \frac{1}{2} \frac{1}{S^2} (\sigma^2 S^2) dt.$$

Therefore

$$d \ln S = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW.$$

Example: $f(X) = e^X$

For $f(X) = e^X$, both first and second derivatives equal e^X. Thus

$$d(e^X) = e^X \left(a + \frac{1}{2}b^2\right) dt + e^X b \, dW.$$

The convexity of the exponential function raises the drift by $b^2/2$.

Itô's Lemma for an Option

Let the option value be $V(S,t)$ and let

$$dS = \mu S dt + \sigma S dW.$$

Then

$$dV = \left(V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} \right) dt + \sigma S V_S dW.$$

This equation is the bridge from the stock process to option pricing.

Meaning of the Derivatives

- V_t measures time decay with the stock price fixed.
- $V_S = \Delta$ measures first-order exposure to the stock price.
- $V_{SS} = \Gamma$ measures curvature of the option value.
- The term $\frac{1}{2}\sigma^2 S^2 \Gamma$ is the Itô correction caused by price uncertainty.

9.8 No-Arbitrage and Replication

Why expected stock return disappears from option value

Law of One Price

- Two portfolios with identical future cash flows must have the same current value.
- If prices differ, traders buy the cheaper portfolio and sell the more expensive one.
- A persistent price difference would create an arbitrage profit without net risk or investment.

A Locally Riskless Portfolio

Hold Δ shares and short one option:

$$\Pi = \Delta S - V.$$

Over a short interval,

$$d\Pi = \Delta dS - dV.$$

Choose Δ so that the dW exposure cancels.

Delta Hedge

The random term in dV is

$$\sigma S V_S dW.$$

The random term in ΔdS is

$$\Delta \sigma S dW.$$

Set

$$\Delta = V_S.$$

The two random terms cancel locally.

Return on the Hedged Portfolio

After choosing $\Delta = V_S$,

$$d\Pi = \left(-V_t - \frac{1}{2}\sigma^2 S^2 V_{SS} \right) dt.$$

The portfolio has no dW term over the instant. No arbitrage therefore requires

$$d\Pi = r\Pi dt.$$

Why μ Disappears

- Both the stock and option contain exposure to the same Brownian shock.
- The delta hedge removes that shock and also removes the stock drift term from the portfolio equation.
- The riskless portfolio earns the risk-free rate.
- Option value therefore depends on r rather than the stock's expected return μ.

Dynamic Replication

- Option delta changes when the stock price or time changes.
- The replicating portfolio must rebalance continuously in the theoretical model.
- Real trading is discrete and costly, so actual hedges retain residual risk.
- Gamma measures how quickly delta changes with the stock price.

9.9 The Black–Scholes Partial Differential Equation

Complete algebra from the hedge to the PDE

Step 1: Stock Dynamics

Start with

$$dS = \mu S dt + \sigma S dW.$$

The stock has expected proportional change μdt and random proportional change σdW.

Step 2: Option Dynamics

Itô's lemma gives

$$dV = \left(V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} \right) dt + \sigma S V_S dW.$$

Step 3: Form the Portfolio

Set

$$\Pi = V_S S - V.$$

Over an instant, hold V_S fixed while computing the portfolio change:

$$d\Pi = V_S dS - dV.$$

Step 4: Cancel the Random Shock

Substitute dS and dV:

$$d\Pi = V_S(\mu S dt + \sigma S dW) - \left[\left(V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right) dt + \sigma S V_S dW\right].$$

The $\mu S V_S dt$ and $\sigma S V_S dW$ terms cancel.

Step 5: Riskless Portfolio Change

After cancellation,

$$d\Pi = -\left(V_t + \frac{1}{2}\sigma^2 S^2 V_{SS}\right) dt.$$

The portfolio value is

$$\Pi = SV_S - V.$$

Step 6: Apply No Arbitrage

A locally riskless self-financing portfolio earns the risk-free rate:

$$-\left(V_t + \frac{1}{2}\sigma^2 S^2 V_{SS}\right) dt = r(SV_S - V)dt.$$

Move every term to the left:

$$\boxed{V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0}.$$

Boundary and Terminal Conditions

The PDE alone does not identify a unique option price. A European call requires

$$V(S, T) = \max(S - K, 0).$$

Useful boundaries are

$$V(0, t) = 0, \qquad V(S, t) \sim S - Ke^{-r(T-t)} \text{ as } S \to \infty.$$

Economic Meaning of the PDE

- V_t captures the passage of time.
- $\frac{1}{2}\sigma^2 S^2 V_{SS}$ prices curvature under uncertainty.
- rSV_S reflects the financing of the stock component.
- $-rV$ reflects the required return on the option value after hedging.

9.10 Solving the Black–Scholes PDE

From the pricing equation to the normal distribution

Change of Variables

Define time to maturity and log moneyness:

$$\tau = \frac{1}{2}\sigma^2(T-t), \qquad x = \ln(S/K).$$

Scale the option value:

$$v(x,\tau) = V(S,t)/K.$$

The logarithm converts the multiplicative stock variable into an additive variable.

Derivative Transformation

Because $x = \ln(S/K)$,

$$SV_S = Kv_x, \qquad S^2 V_{SS} = K(v_{xx} - v_x).$$

Because $\tau = \frac{1}{2}\sigma^2(T - t)$,

$$V_t = -\frac{1}{2}\sigma^2 K v_\tau.$$

Dimensionless PDE

Let

$$k = \frac{2r}{\sigma^2}.$$

Substitution into the Black–Scholes PDE gives

$$v_\tau = v_{xx} + (k-1)v_x - kv.$$

This equation still contains a first derivative and a level term.

Remove the Extra Terms

Set

$$v(x,\tau) = e^{ax+b\tau}u(x,\tau).$$

Choose

$$a = \frac{1-k}{2}, \qquad b = -\frac{(k+1)^2}{4}.$$

These constants cancel the u_x and u terms.

The Heat Equation

After the transformation,

$$u_\tau = u_{xx}.$$

The Black–Scholes pricing problem has become the classical heat equation. Diffusion of heat and diffusion of probability share the same mathematical structure.

Heat Kernel

The heat equation solution averages the initial condition with a Gaussian kernel:

$$u(x,\tau) = \frac{1}{\sqrt{4\pi\tau}} \int_{-\infty}^{\infty} u(y,0) \exp\left[-\frac{(x-y)^2}{4\tau}\right] dy.$$

This Gaussian integral explains why the normal CDF enters the final option formula.

Terminal Payoff under the Transformation

At maturity, $\tau = 0$ and

$$v(x, 0) = \max(e^x - 1, 0).$$

The payoff is positive only when $x > 0$, which means $S > K$. The heat-kernel integral therefore splits at the exercise boundary.

An Equivalent Risk-Neutral Route

The PDE solution equals a discounted expectation under a risk-neutral probability measure:

$$V_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[\text{payoff} \mid \mathcal{F}_t].$$

Under $\mathbb{Q}$,

$$dS = rSdt + \sigma SdW^{\mathbb{Q}}.$$

The stock drift changes from μ to r for pricing.

9.11 The Black–Scholes Formula

Evaluation of the discounted payoff integral

Risk-Neutral Distribution of S_T

Let $\tau = T - t$. Under the risk-neutral measure,

$$\ln S_T \sim N\left(\ln S_t + \left(r - \frac{1}{2}\sigma^2\right)\tau,\ \sigma^2\tau\right).$$

Equivalently,

$$S_T = S_t \exp\left[\left(r - \frac{1}{2}\sigma^2\right)\tau + \sigma\sqrt{\tau}Z\right].$$

Split the Call Payoff

A European call payoff is

$$(S_T - K)^+ = S_T \mathbf{1}_{\{S_T > K\}} - K \mathbf{1}_{\{S_T > K\}}.$$

Therefore

$$C_t = e^{-r\tau} \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T > K\}}] - K e^{-r\tau} \overset{\mathbb{Q}}{\Pr}(S_T > K).$$

Exercise Probability and d_2

The event $S_T > K$ becomes

$$Z > \frac{\ln(K/S_t) - (r - \sigma^2/2)\tau}{\sigma\sqrt{\tau}}.$$

Using normal symmetry,

$$\overset{\mathbb{Q}}{\Pr}(S_T > K) = N(d_2),$$

where

$$d_2 = \frac{\ln(S_t/K) + (r - \sigma^2/2)\tau}{\sigma\sqrt{\tau}}.$$

Stock-Weighted Term and d_1

Evaluating the truncated lognormal expectation gives

$$e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T>K\}}] = S_t N(d_1),$$

where

$$d_1 = \frac{\ln(S_t/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}} = d_2 + \sigma\sqrt{\tau}.$$

European Call Formula

Combining the two payoff components gives

$$\boxed{C_t = S_t N(d_1) - K e^{-r\tau} N(d_2)}.$$

The first term is the option-adjusted value of receiving the stock. The second term is the option-adjusted present value of paying the strike.

European Put Formula

Put–call parity gives

$$C_t - P_t = S_t - Ke^{-r\tau}.$$

Therefore

$$\boxed{P_t = Ke^{-r\tau}N(-d_2) - S_tN(-d_1)}.$$

Interpretation of $N(d_1)$ and $N(d_2)$

- $N(d_1)$ is the call delta for a non-dividend-paying stock.
- $N(d_2)$ is the risk-neutral probability that the call finishes in the money.
- The two numbers differ because the first expectation weights outcomes by the future stock price.
- $d_1 = d_2 + \sigma\sqrt{\tau}$.

Why the Formula Contains Two Uses of e

- e^{-rT} discounts the future strike payment continuously.
- The lognormal stock model comes from exponentiating a normal log return.
- Both appearances trace back to continuous proportional growth.
- The normal density itself also contains an exponential term.

9.12 Interpretation, Extensions, and Limits

What the model assumes and how practitioners adapt it

Black–Scholes Assumptions

- The stock follows geometric Brownian motion with constant volatility.
- Markets permit continuous trading without taxes or transaction costs.
- Investors can borrow and lend at the risk-free rate.
- Short selling is possible and securities are divisible.
- The option is European and the basic stock pays no dividend.

Dividend Yield Extension

For a continuous dividend yield q, risk-neutral stock dynamics become

$$dS = (r - q)Sdt + \sigma SdW^{\mathbb{Q}}.$$

The call price becomes

$$C = Se^{-q\tau}N(d_1) - Ke^{-r\tau}N(d_2),$$

with

$$d_1 = \frac{\ln(S/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}.$$

The Greeks

Greek	Derivative	Main exposure
Delta	V_S	stock-price sensitivity
Gamma	V_{SS}	change in delta
Theta	V_t	passage of time
Vega	V_σ	volatility sensitivity
Rho	V_r	interest-rate sensitivity

Volatility and Option Value

- A call holder has limited downside and retains upside.
- Greater volatility spreads the distribution of future prices.
- The favorable tail becomes more valuable while the payoff floor remains zero.
- Both call and put values generally rise with volatility.

Implied Volatility

Market price can be inserted into the Black–Scholes formula and solved for σ:

$$C_{market} = BS(S, K, r, \tau, \sigma_{impl}).$$

Implied volatility summarizes the volatility level consistent with the observed option price.

Volatility Smile

- Constant volatility predicts the same volatility for all strikes and maturities.
- Observed implied volatility commonly varies across strike and maturity.
- The pattern reflects skewness, heavy tails, jumps, supply and demand, and market risk premiums.
- The smile shows where the benchmark model misses market behavior.

Global Financial Strategy Applications

- Currency options protect uncertain foreign-currency cash flows while preserving favorable exchange-rate movements.
- Equity can be viewed as an option on firm assets when debt is outstanding.
- Investment timing creates real options when management can wait for country or regulatory information.
- Option-based compensation affects dilution and managerial risk incentives.

Model Risk

- Volatility changes over time and cannot be observed directly.
- Prices jump after major announcements or policy changes.
- Continuous hedging is impossible in practice.
- Liquidity, transaction costs, and funding constraints become important during stress.
- The formula is a benchmark that requires judgment and stress testing.

Full Derivation Map

1. Continuous proportional growth produces e and logarithms.
2. Many small shocks motivate Brownian motion and normal increments.
3. Exponentiating normal log returns produces positive lognormal prices.
4. Brownian quadratic variation produces Itô's correction term.
5. Delta hedging removes the random shock and the unknown stock drift.
6. No arbitrage produces the PDE.
7. Gaussian integration produces $N(d_1)$ and $N(d_2)$.

9.P 30-Minute Practice

Foundations, derivation, and interpretation

Practice Instructions

- Work individually for 20 minutes.
- Compare derivations with a partner for 5 minutes.
- Use the final 5 minutes to discuss economic interpretation.
- Show the order of each small term when applying Itô calculus.

Practice 1: Continuous Compounding

A principal of \$1,000 earns a continuously compounded annual rate of 5% for three years. Calculate the future value and explain why e appears.

Practice 2: Log Return

A stock rises from \$80 to \$92. Compute the continuously compounded return and compare it with the simple return.

Practice 3: Brownian Increment

Let $dt = 0.01$ year and $dW = \sqrt{dt}Z$. If $Z = -0.6$, compute dW. Compare the sizes of dt, dW, $(dW)^2$, and $dt\,dW$.

Practice 4: One Itô Step

If $dX = 2dt + 3dW$, use Itô's lemma to find $d(X^2)$.

Practice 5: Log Stock Price

Starting from $dS = \mu S dt + \sigma S dW$, derive $d \ln S$ and identify the Itô correction.

Practice 6: Delta Hedge

An option has delta 0.65. Construct a portfolio using shares and one short option that eliminates the instantaneous dW exposure.

Practice 7: PDE Algebra

Starting with $\Pi = V_S S - V$, reproduce the algebra leading to the Black–Scholes PDE.

Practice 8: Compute d_1 and d_2

Let $S = 100$, $K = 100$, $r = 4\%$, $\sigma = 20\%$, and $\tau = 1$. Compute d_1 and d_2.

Practice 9: European Call

Using $N(0.30) = 0.6179$ and $N(0.10) = 0.5398$, calculate the call value for the data in Practice 8.

Practice 10: Assumption Diagnosis

A currency suddenly jumps 12% after a policy announcement. Identify the Black–Scholes assumptions challenged by this event and explain the likely hedging consequence.

9.S Solutions and Debrief

Detailed answers

Solution 1

$$FV = 1{,}000e^{0.05(3)} = \boxed{1{,}161.83}.$$

e appears because continuous compounding is the limit of increasingly frequent discrete compounding.

Solution 2

$$R_{log} = \ln(92/80) = \boxed{13.976\%}.$$

$$R_{simple} = 92/80 - 1 = \boxed{15\%}.$$

The relationship is $R_{log} = \ln(1 + R_{simple})$.

Solution 3

$$dW = \sqrt{0.01}(-0.6) = \boxed{-0.06}.$$

Orders: $dt = 0.01$, $dW = -0.06$, $(dW)^2 = 0.0036$ for this realization, and $dt\,dW = -0.0006$. In stochastic-order reasoning, $(dW)^2$ accumulates like dt, while $dt\,dW$ vanishes.

Solution 4

For $f(X) = X^2$, $f_X = 2X$ and $f_{XX} = 2$:

$$d(X^2) = 2X(2dt + 3dW) + \frac{1}{2}(2)(3^2)dt.$$

$$\boxed{d(X^2) = (4X + 9)dt + 6XdW}.$$

Solution 5

For $f(S) = \ln S$, $f_S = 1/S$ and $f_{SS} = -1/S^2$:

$$d \ln S = \frac{1}{S} dS + \frac{1}{2} \left(-\frac{1}{S^2} \right) (\sigma^2 S^2) dt.$$

$$\boxed{d \ln S = (\mu - \sigma^2/2) dt + \sigma dW}.$$

Solution 6

Short one option and buy 0.65 shares:

$$\Pi = 0.65S - V.$$

- The stock contributes $0.65\sigma S dW$.
- The short option contributes $-V_S \sigma S dW = -0.65\sigma S dW$.
- The random terms cancel locally.

Solution 7

$$d\Pi = V_S dS - dV = -\left(V_t + \frac{1}{2}\sigma^2 S^2 V_{SS}\right) dt.$$

No arbitrage requires $d\Pi = r(V_S S - V)dt$. Rearranging,

$$\boxed{V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0}.$$

Solution 8

$$d_1 = \frac{\ln(1) + (0.04 + 0.20^2/2)(1)}{0.20} = \boxed{0.30}.$$

$$d_2 = d_1 - 0.20 = \boxed{0.10}.$$

Solution 9

$$C = 100N(0.30) - 100e^{-0.04}N(0.10).$$

$$C = 100(0.6179) - 100(0.96079)(0.5398) \approx \boxed{9.93}.$$

Solution 10

A sudden jump violates continuous sample paths and challenges constant volatility and normal log-return assumptions. Delta computed immediately before the announcement cannot eliminate jump risk. The hedge may require jump or stochastic-volatility models, wider stress scenarios, and explicit liquidity reserves.

Chapter 9 Takeaways

- Continuous growth explains the exponential and logarithmic terms.
- Brownian scaling explains dW, quadratic variation, and the Itô correction.
- Delta hedging and no arbitrage produce the Black–Scholes PDE.
- The normal CDF enters when the lognormal payoff integral is evaluated.
- The model provides a benchmark, while actual risk management must address jumps, changing volatility, costs, and liquidity.